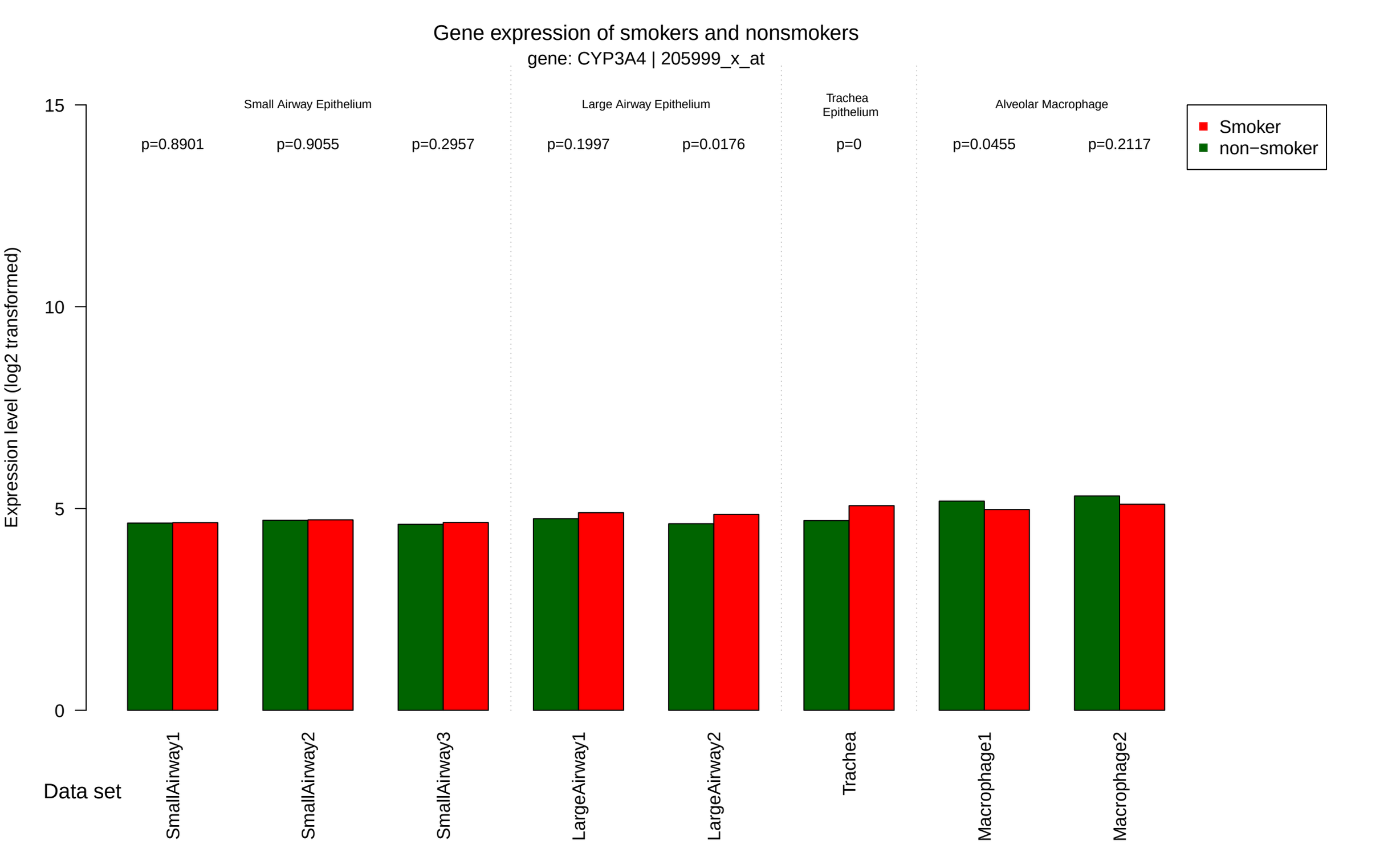
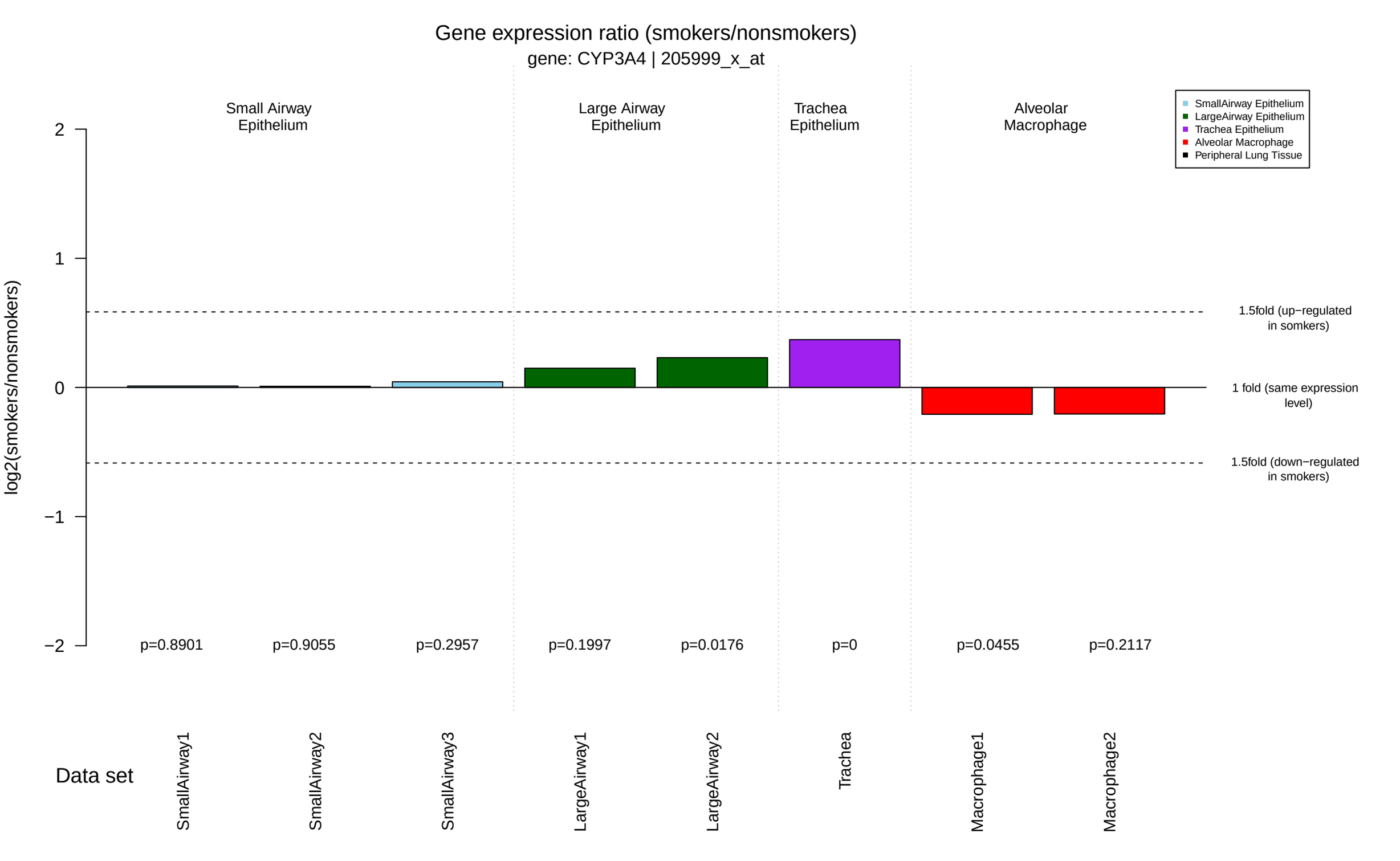
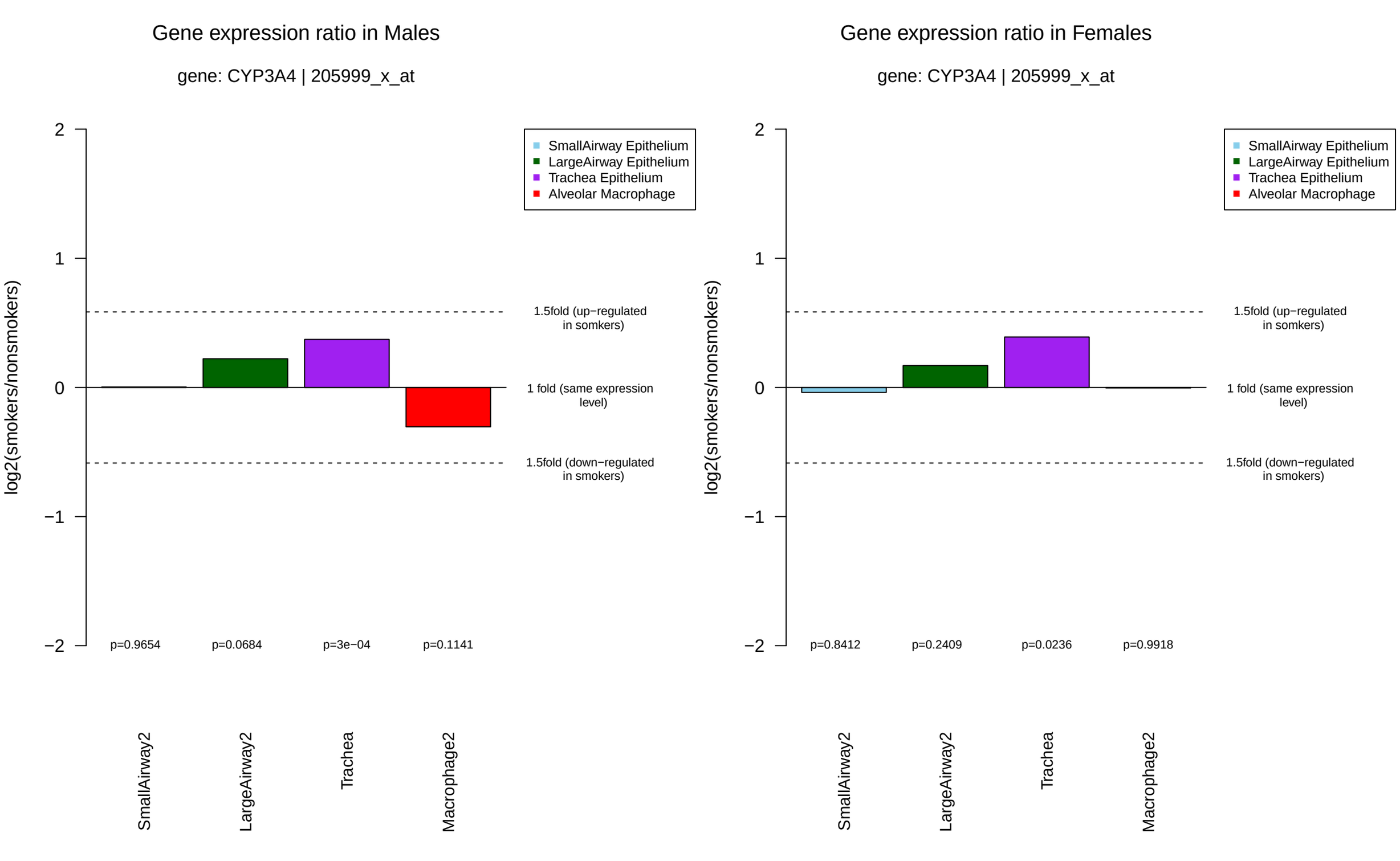
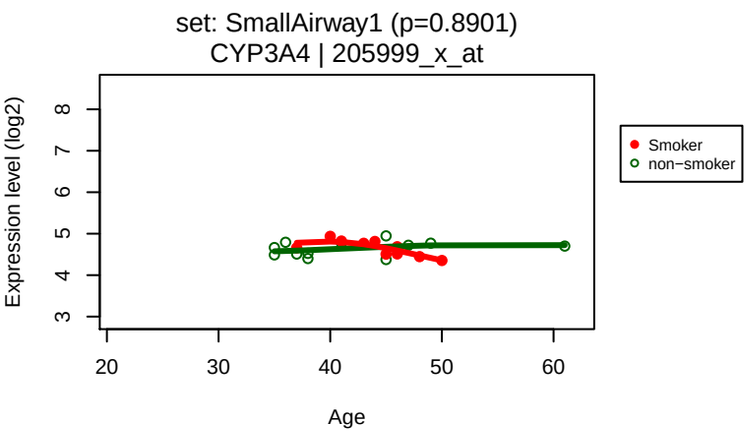
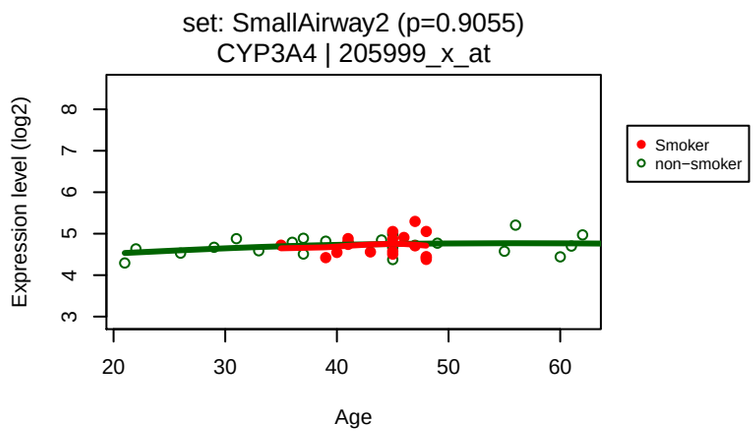
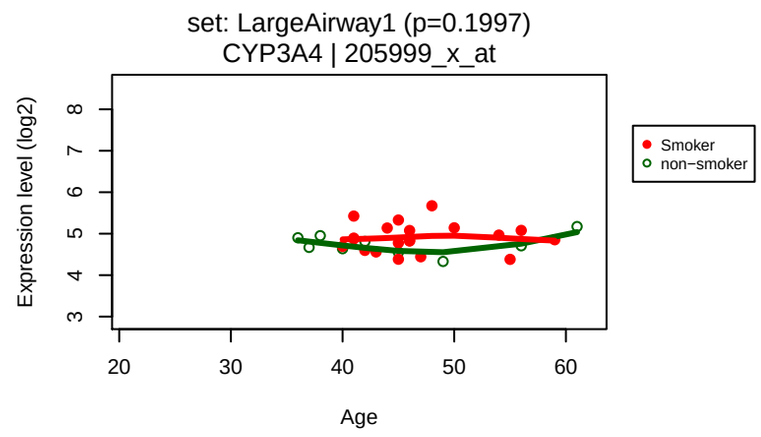

Gene expression of smokers and nonsmokers
gene: CYP3A4 | 205999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8901
p=0.9055
p=0.2957
p=0.1997
p=0.0176
p=0
p=0.0455
p=0.2117
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

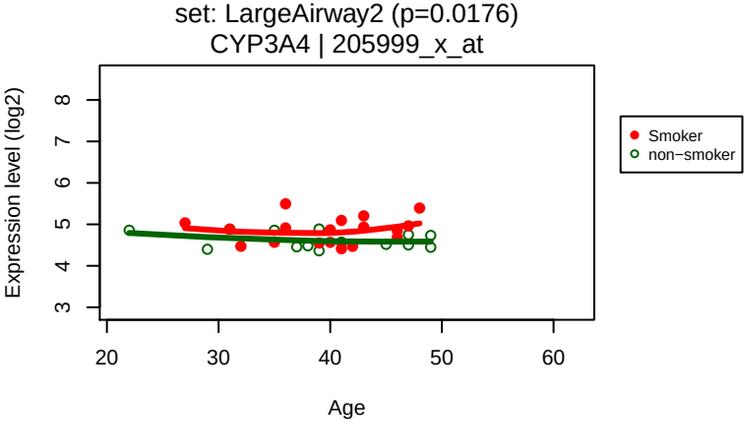

Gene expression ratio (smokers/nonsmokers)
gene: CYP3A4 | 205999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8901
p=0.9055
p=0.2957
p=0.1997
p=0.0176
p=0
p=0.0455
p=0.2117
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

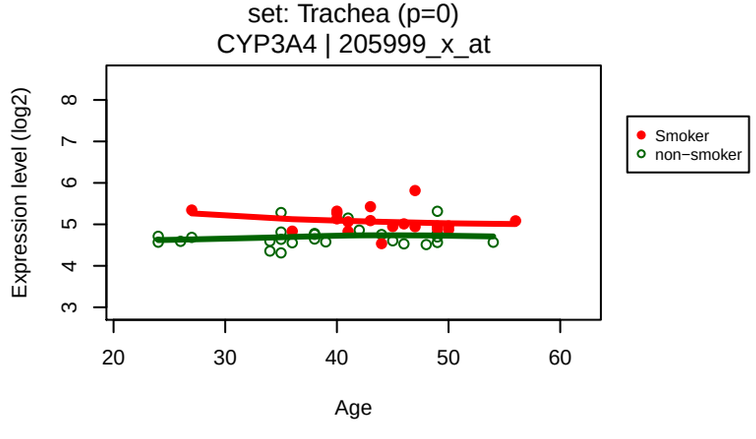

Gene expression ratio in Males
gene: CYP3A4 | 205999_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9654
p=0.0684
p=3e−04
p=0.1141
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP3A4 | 205999_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8412
p=0.2409
p=0.0236
p=0.9918
SmallAirway2
LargeAirway2
Trachea
Macrophage2

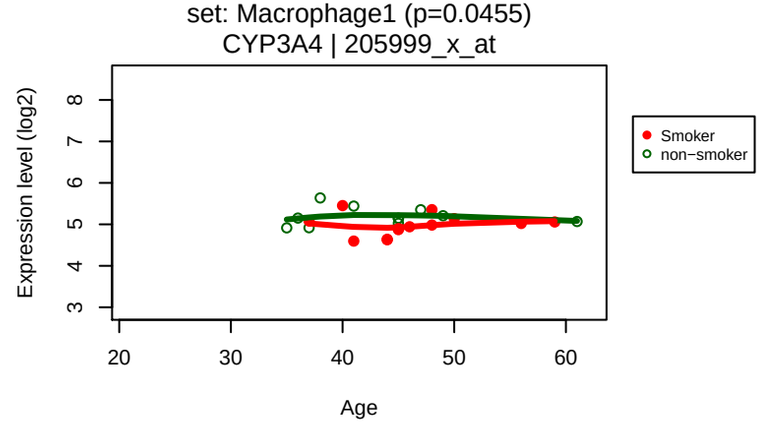

set: SmallAirway1 (p=0.8901)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker

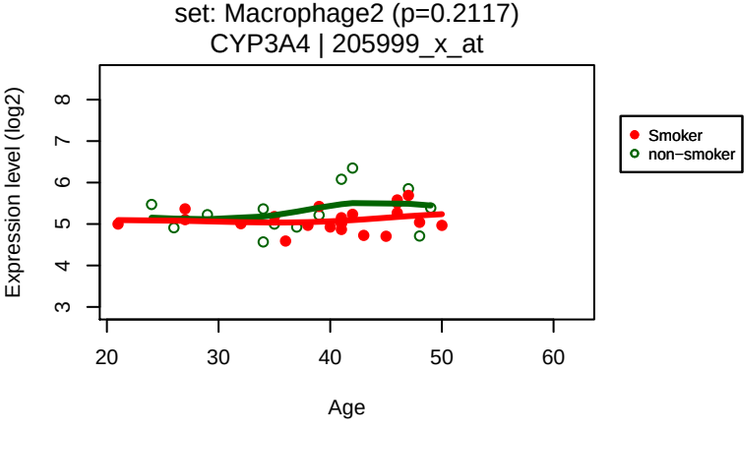

set: SmallAirway2 (p=0.9055)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker

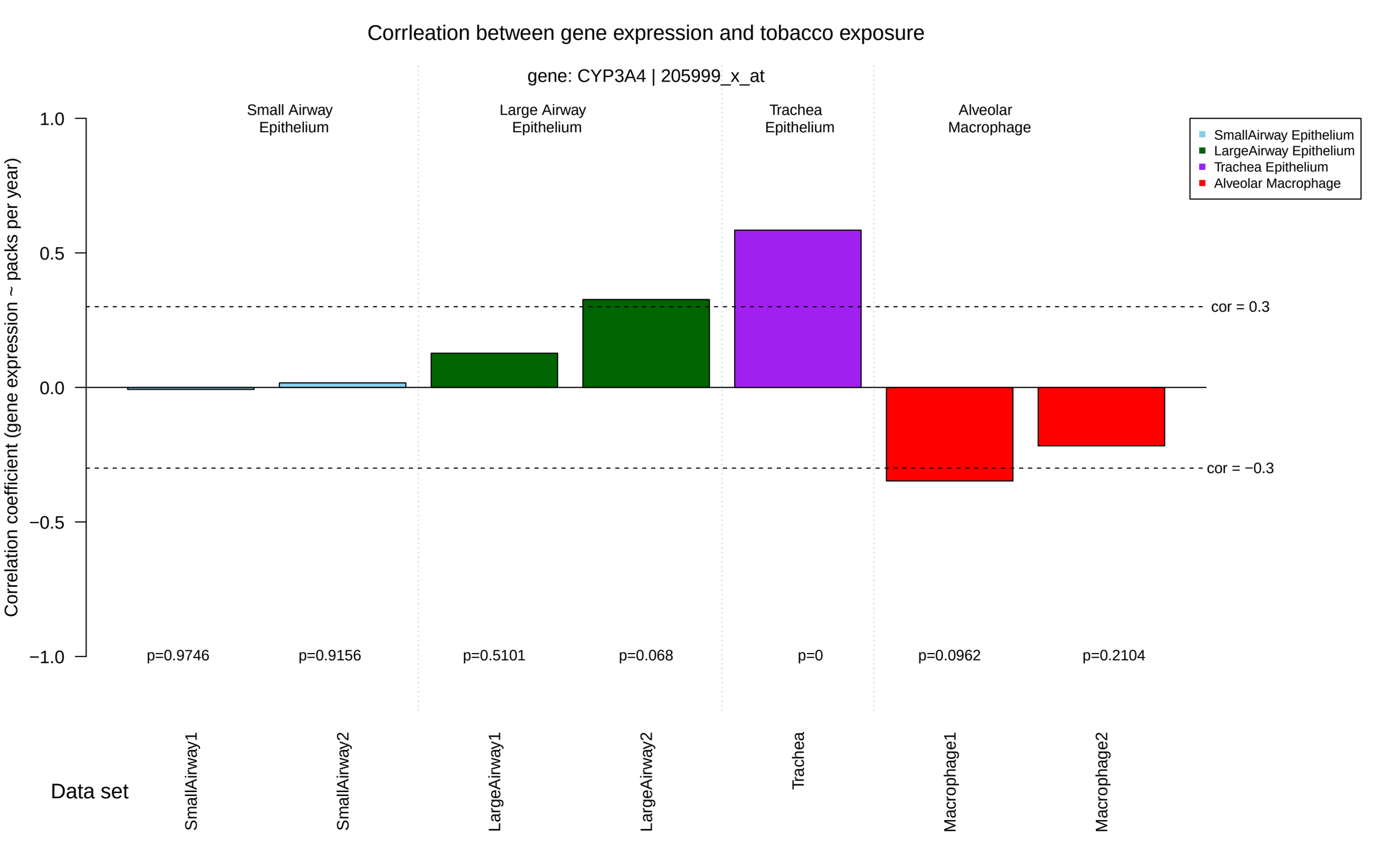

set: LargeAirway1 (p=0.1997)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0176)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0455)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2117)
CYP3A4 | 205999_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP3A4 | 205999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9746
p=0.9156
p=0.5101
p=0.068
p=0
p=0.0962
p=0.2104
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2